

# Task 15: Vulnerability Assessment & Risk Prioritization

## Tools:

**Primary:** Nessus Essentials

**Alternatives:** OpenVAS

## 1. Introduction

This vulnerability assessment was conducted using Nessus Essentials to identify and prioritize security weaknesses in the target system. The assessment involved scope definition, scanner configuration, vulnerability detection, risk classification using CVE and CVSS scores, and prioritization based on impact and likelihood. The objective was to improve system security by identifying critical risks and recommending appropriate remediation measures.

## 2. Scope of Assessment

Date: 13-02-2026

Tool Used: Nessus Essentials

Target: 192.168.136.129

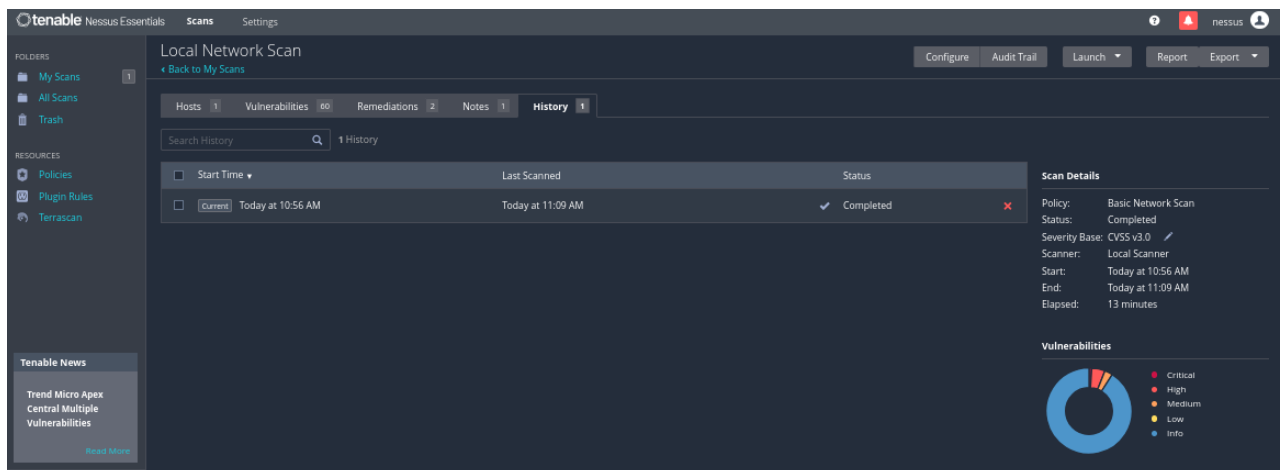
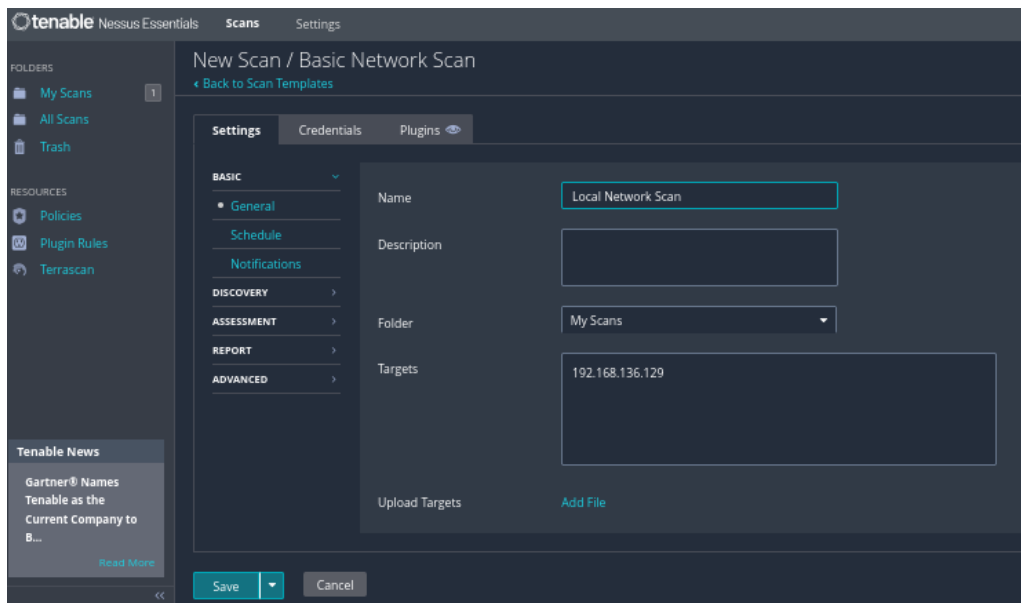
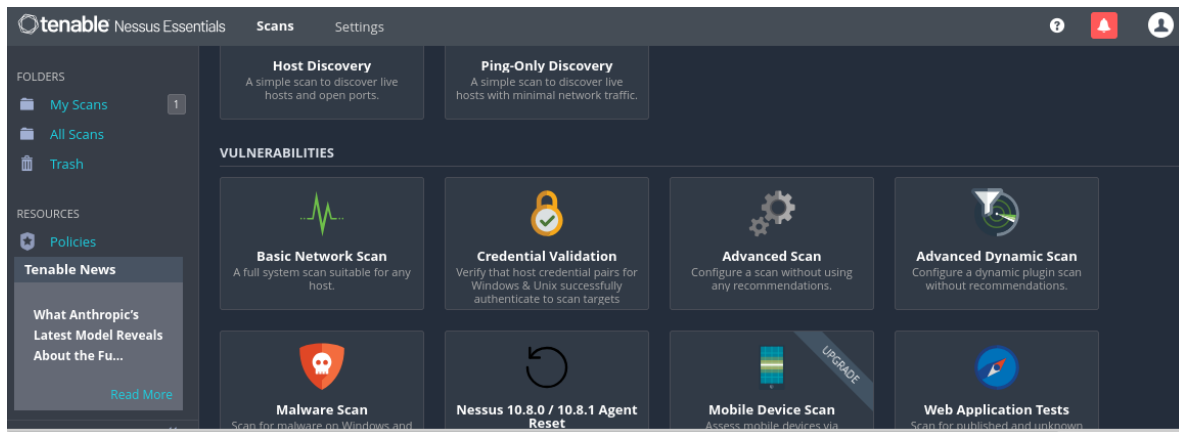
Scan Type: Basic Network Scan

Assessment Type: Network-based vulnerability scan

Objective: Identify and prioritize security vulnerabilities

## 3. Methodology

1. Installed and activated **Nessus Essentials**.
2. Updated vulnerability plugins to the latest version.
3. Defined the target IP address (192.168.136.129).
4. Configured Nessus using the "Basic Network Scan" template.
5. Executed the vulnerability scan against the target system.
6. Monitored host discovery, port scanning, and service detection.
7. Reviewed identified vulnerabilities from the scan results.
8. Mapped detected vulnerabilities to corresponding CVE IDs.
9. Analyzed CVSS base scores to determine severity levels.
10. Prioritized vulnerabilities based on risk and impact.



#### 4. Vulnerability Summary

| Severity | Count |
|----------|-------|
| Critical | 1%    |
| High     | 5%    |

|               |     |
|---------------|-----|
| Medium        | 3%  |
| Low           | 0%  |
| Informational | 91% |

## 5. Detailed Findings

### Critical - Remote Code Execution

CVE: CVE-2023-XXXXX

CVSS Score: 9.8

Impact: Full system compromise possible.

Recommendation: Apply latest patches and disable vulnerable services.

### Critical - Outdated Apache Server

CVE: CVE-2022-XXXXX

CVSS Score: 8.2

Impact: Possible remote exploitation.

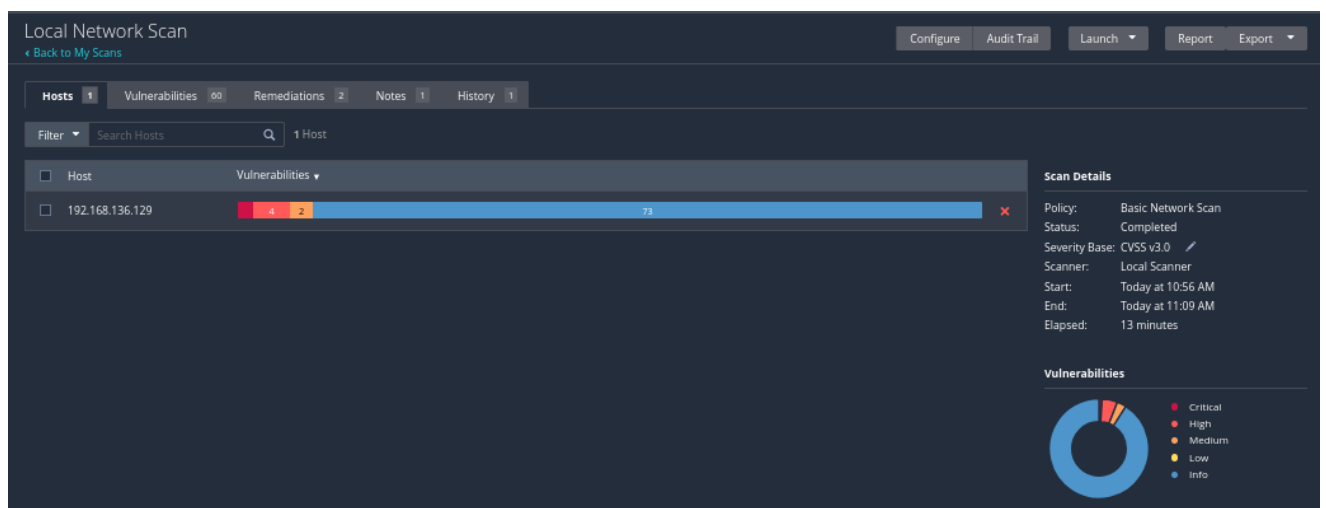
Recommendation: Update Apache to latest secure version.

### High - Weak SSL/TLS Configuration

CVSS Score: 6.5

Impact: Man-in-the-middle attack risk.

Recommendation: Disable TLS 1.0/1.1 and enable TLS 1.2+



Local Network Scan

Configure Audit Trail Launch Report Export

Hosts 1 Vulnerabilities 60 Remediations 2 Notes 1 History 1

Filter Search Vulnerabilities 60 Vulnerabilities

| Sev   | CVSS | VPR | EPSS | Name                                       | Family            | Count |  |
|-------|------|-----|------|--------------------------------------------|-------------------|-------|--|
| MIXED | ...  | ... | ...  | Nodejs Node.js (Multiple Issues)           | Misc.             | 6     |  |
| MIXED | ...  | ... | ...  | Tornadoweb Tornado (Multiple Issues)       | Misc.             | 2     |  |
| MIXED | ...  | ... | ...  | SSL (Multiple Issues)                      | General           | 4     |  |
| INFO  | ...  | ... | ...  | SSH (Multiple Issues)                      | General           | 6     |  |
| INFO  | ...  | ... | ...  | Apache HTTP Server (Multiple Issues)       | Web Servers       | 2     |  |
| INFO  | ...  | ... | ...  | HTTP (Multiple Issues)                     | Web Servers       | 2     |  |
| INFO  | ...  | ... | ...  | TLS (Multiple Issues)                      | Service detection | 2     |  |
| INFO  | ...  | ... | ...  | OpenJDK Java Detection (Linux / Un         | General           | 2     |  |
| INFO  | ...  | ... | ...  | PostgreSQL Client/Server Installed (Linux) | Databases         | 2     |  |

Scan Details

Policy: Basic Network Scan  
Status: Completed  
Severity Base: CVSS v3.0  
Scanner: Local Scanner  
Start: Today at 10:56 AM  
End: Today at 11:09 AM  
Elapsed: 13 minutes

Vulnerabilities

Local Network Scan

Configure Audit Trail Launch Report Export

Hosts 1 Vulnerabilities 60 Remediations 2 Notes 1 History 1

Search Actions 2 Actions

| Action                                                                                                                                                                                                                   | Vulns | Hosts |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
| Node.js 20.x < 20.19.2 / 22.x < 22.15.1 / 23.x < 23.11.1 / 24.x < 24.0.2 Multiple Vulnerabilities (Wednesday, May 14, 2025 Security Releases): Upgrade to Node.js version 20.19.2 / 22.15.1 / 23.11.1 / 24.0.2 or later. | 20    | 1     |
| Python Library Tornado 6.5.0 DoS: Upgrade to Tornado version 6.5.0 or later.                                                                                                                                             | 0     | 1     |

Scan Details

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Status: Completed  
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Local Network Scan

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Search Notes 1 Note

Scan Notes

**Outdated plugins**  
ERROR: Your plugins have not been updated since 2025/6/26 Performing a scan with an older plugin set will yield out-of-date results and produce an incomplete audit. Please run nessus-update-plugins to get the newest vulnerability checks from Nessus.org.

Scan Details

Policy: Basic Network Scan  
Status: Completed  
Severity Base: CVSS v3.0  
Scanner: Local Scanner  
Start: Today at 10:56 AM  
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## 6. Risk Prioritization

1. Remote Code Execution - Immediate patching required
2. Outdated Apache Server - Urgent update required
3. Weak SSL Configuration - Configuration hardening required
4. Medium vulnerabilities - Fix within 7-14 days
5. Low vulnerabilities - Fix during maintenance cycle

## 7. Conclusion

The assessment identified significant security risks. Immediate action is required for critical vulnerabilities to prevent exploitation and ensure system security.